

COMPUTER VISION • INSTANCE SEGMENTATION

Utility Pole Instance Segmentation

End-to-end pipeline using MMDetection & Mask R-CNN
for automated pole detection and mask generation

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Problem Statement

Manual Inspection Risk

Field engineers manually inspect thousands of poles — slow, dangerous, and error-prone.

No Spatial Precision

Conventional methods lack pixel-level localization of pole boundaries and components.

No Height Data

Estimating pole height from images requires structured depth cues — impossible without AI.



Raw Input Image

Solution Overview

An automated computer vision pipeline that detects utility poles in images, generates precise instance masks, and enables downstream analysis like height prediction.



3

Image Types Processed

Aerial, Street & Close-up

12

Team Members Led

Full delivery as Project Lead

3D

Dataset Created

Custom pole 3D dataset

Tech Stack

MMDetection

DETECTION FRAMEWORK

Industry-standard open-source framework for object detection & segmentation

Mask R-CNN

CORE MODEL

Two-stage instance segmentation model with RPN + mask branch

Python

LANGUAGE

OOP-based modular code architecture for clean integration & maintenance

OpenCV

IMAGE PROCESSING

Real-time image manipulation, visualization, and pre/post-processing

JSON Parser

DATA HANDLING

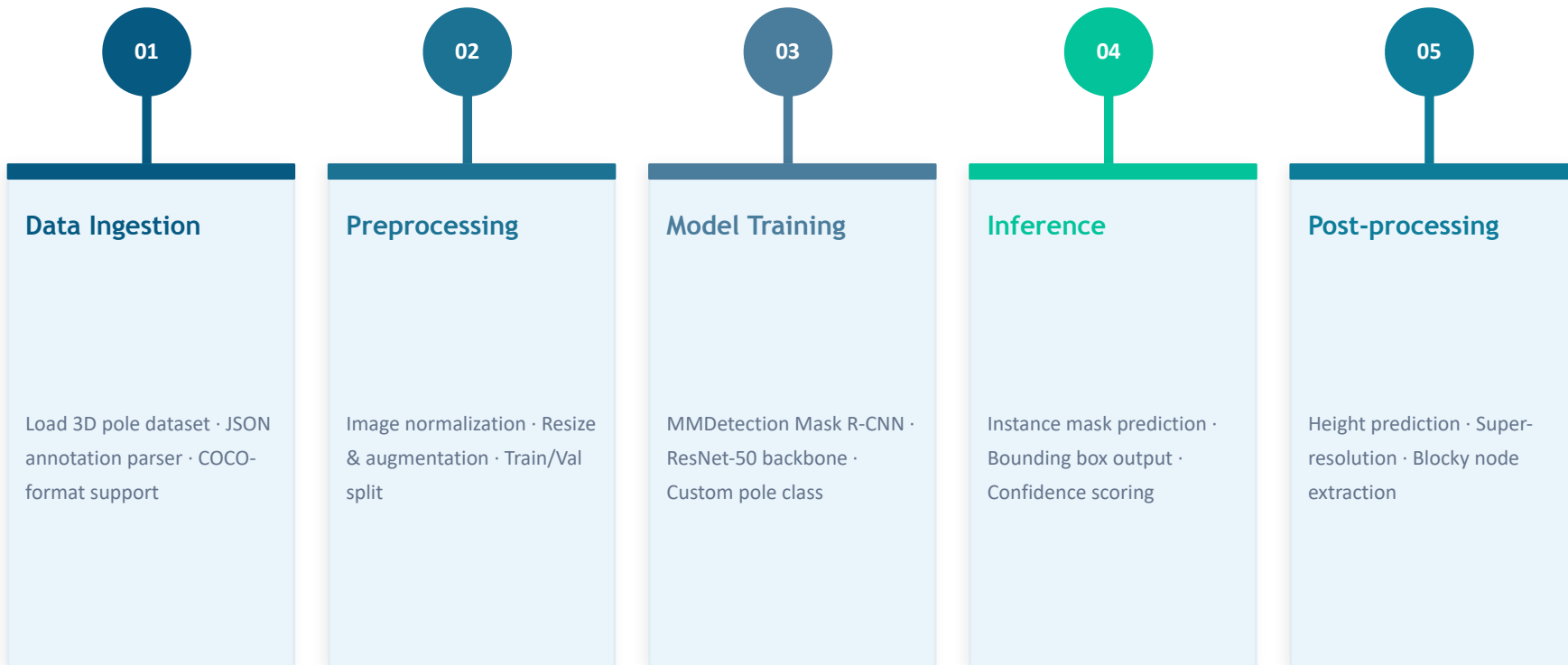
Custom-built parser for COCO-format annotation loading & dataset management

Docker

DEPLOYMENT

Containerized inference pipeline for reproducible, portable deployment

Pipeline Architecture

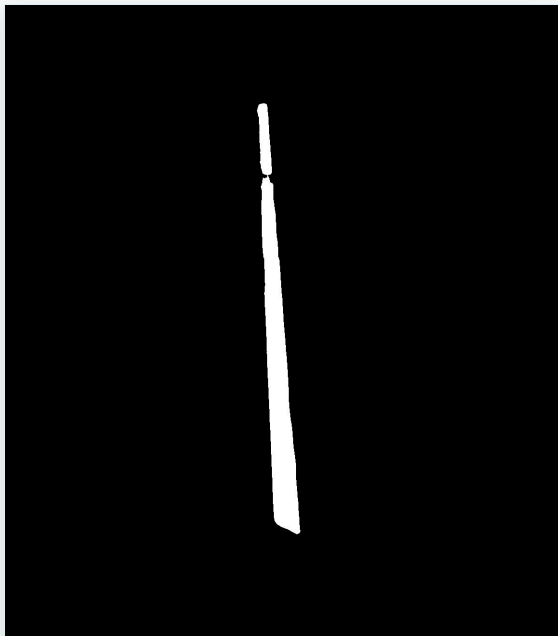


Segmentation Results



01 Original Input

Raw field photograph



02 Binary Mask

Predicted segmentation mask



03 Mask Overlay

Instance mask on original image

Key Features Implemented



Instance Segmentation

- MMDetection Mask R-CNN
- Per-instance pixel masks
- Multi-pole detection in one shot
- COCO-format annotation pipeline



Height Prediction

- Pixel mask height extraction
- Camera metadata integration
- Trained ML regression model
- Real-world scale calibration



Super Resolution

- Enhanced image upscaling
- Detail recovery for thin poles
- Improves mask edge accuracy
- Pre-inference enhancement

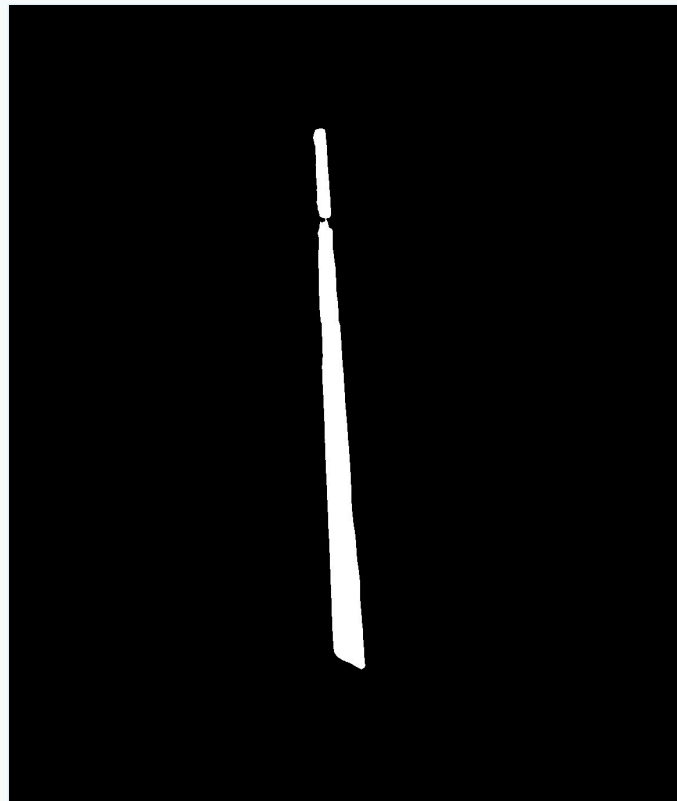


Blocky Node Creation

- 3D structural node extraction
- JSON-based node graph output
- Enables 3D reconstruction
- Custom dataset generation

Dataset & Annotation Strategy

Dataset Type	Custom 3D Pole Dataset
Annotation Format	COCO JSON (polygons + RLE)
Image Sources	Aerial, Street-level, Close-up
Classes	Utility Pole (single custom class)
Annotation Tool	Custom JSON Parser (built in-house)
Augmentation	Flip, rotate, scale, brightness



Binary segmentation mask

System Architecture



Challenges & Solutions

Challenge

Thin pole structures hard to segment

No existing labeled pole dataset

Height estimation from 2D images

Multi-instance poles in one frame

Solution

Super-resolution pre-processing to enhance fine edges before inference

Built custom 3D pole dataset + JSON parser from scratch with team of 12

Pixel mask height combined with camera metadata fed into regression model

Instance segmentation (vs semantic) assigns unique ID & mask per pole

Key Achievements



Led Team of 12

Successfully coordinated a 12-member team end-to-end from dataset creation to final delivery



On-time Delivery

Met all client specifications and deadlines despite complex multi-component requirements



Custom Dataset Built

Created a first-of-its-kind 3D utility pole dataset with full COCO-format annotation pipeline



Height Prediction Module

Engineered novel height estimation from pixel masks and camera metadata — no depth sensor required



Super-Resolution Pipeline

Integrated super-res upscaling as pre-processing step improving thin-pole segmentation accuracy



3D Node Extraction

Built blocky node creation system enabling 3D structural reconstruction from 2D imagery

Thank You

Questions & Discussion

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